Somya Ranjan Sahu

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.53
Vyomayana Samstha Vidyalaya	Class XII (Central Board of Sencondary Education)	2020	82.60%
Vyomayana Samstha Vidyalaya	Class X (Central Board of Sencondary Education)	2018	88.20%

INTERNSHIPS

Indian Oil Corporation Limited

Summer Intern, Paradip, Odisha

- Analysed real-time reactor process data for an Ethylene Oxide unit and applied **regression analysis** to evaluate catalyst activity and selectivity against reference benchmarks, supporting ongoing process-performance monitoring efforts.

- Studied refinery unit operations (distillation, catalytic cracking, hydroprocessing, reforming, etc.) to understand **process optimization, quality control and safety practices** implemented across the plant.

Hindustan Aeronautics Limited

Summer Intern, Sunabeda, Odisha

- Understood elemental and alloy characterization using **OES, XRF, CS Analyzer and Flame Photometer** for precision material quality verification of aerospace grade components and assemblies.

- Evaluated material testing and quality control protocols supporting dimensional and compositional accuracy required in high precision aerospace fabrication.

National Aluminium Company Limited

Summer Intern, Damanjodi, Odisha

- Studied the **Bayer process** for alumina production end to end (crushing, digestion, clarification, precipitation, calcination), gaining exposure to large scale continuous process plant operations.

- Reviewed environmental management practices including **red-mud (process residue) management**, an important part of the plant's overall sustainability protocol.

PROJECTS

Linking Rheological Properties of Polymer Solutions to Electrospinning and Fibre Morphology

Masters Research Project *(Electrospining/ Optical Rheometer/ Fluoroscent Microscope/ MATLAB)*

- Employed **extensional rheometry** on **PEO** polymer solutions via Dripping-onto-Substrate (DoS) filament thinning experiments and high speed imaging to characterize a broad range of rheological properties that govern fibre forming behaviour.

- Used the **pendant drop method** to measure surface tension, **optical rheometer** to measure viscosity, enabling a comprehensive characterization of fluid rheological behaviour alongside extensional viscosity and relaxation response, strengthening the overall analysis.

- Built a dimensionless regime-mapping framework (Electric Capillary number, Weber number, Ohnesorge number and

Deborah number) linking fluid, electrical and process parameters to the observed fibre forming behaviour patterns.

- Applied **MATLAB** based image processing to extract filament geometry and quantify thinning dynamics from high speed video data.

CFD Analysis of Pulsatile Non-Newtonian Blood Flow Through Stenosed Arteries

(ANSYS Fluent | SpaceClaim | Carreau viscosity model)

- Built 3D arterial geometries in **ANSYS SpaceClaim** for 25% and 75% stenosis severities and generated structured, quality-verified meshes in **ANSYS Meshing** (element quality 0.33-0.99, aspect ratio <5).

- Modelled blood as a non-Newtonian, shear-thinning fluid using the **Carreau viscosity model** and implemented a custom **C-based UDF** for a pulsatile inlet velocity profile in **ANSYS Fluent**.

- Quantified flow behaviour across severities, showing 75% stenosis produced **~8x velocity amplification** and wall-shear-stress peaks of ~400-450 Pa versus ~20-30 Pa at 25% stenosis.

Process Design & Simulation of Methyl Salicylate Production

(Aspen Plus| Unit Operations| Techno-economic analysis)

- Developed open-loop and closed-loop **Aspen Plus** flowsheets for an acid catalyzed esterification process, modelling reaction kinetics and UNIQUAC based VLE/LLE thermodynamics across a reactive distillation column combining reaction and separation along with other units and utilities.

- Evaluated recycle design's effect on raw material use, utility duty and product purity, achieving **>95% product purity**, cutting the payout period from ~**9 to ~3.7 years** through closed-loop material recycling and reuse.

- Conducted **techno-economic analysis** (capital cost, utility cost, stream costing) via Aspen Process Economic Analyzer to compare the two configurations on profitability.

Green Synthesis & Characterization of Silver Nanoparticles

Undergraduate Research Project ***(UV-Vis spectroscopy/ FTIR/ Bacterial Culture/ Well-Diffusion Method)***

- Synthesized silver nanoparticles via green routes using **Neem** and **Aloe vera** extracts, studying the influence of synthesis time and temperature on nanoparticle formation.

- Characterized synthesized nanoparticles using **UV-Vis spectroscopy** and **FTIR**.

- Studied the **antibacterial effect** of the synthesized Neem and Aloe vera derived silver nanoparticles against test bacterial strains, linking synthesis route to functional biological activity.

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RELEVANT COURSES

Advanced Numerical Methods, Process Engineering Lab, Advanced Process Control, Computational Fluid Dynamics, Machine Learning for Process Systems Engineering, Advanced Transport Phenomena, CFD Lab, Polymer Science and Engineering

SKILLS

Hard Skills

- **Process Simulation:** Aspen Plus, ChemCAD, DWSIM, flowsheet development
- **CFD & Fluid Mechanics:** ANSYS Fluent, ANSYS SpaceClaim, ANSYS Meshing
- **Laboratory Equipments:** UV-Vis spectroscopy, FTIR, Fluorescent Microscope, Optical Rheometer, Electrospinning,

High speed camera, Imaging

- **Tools:** MATLAB, Python, Origin, Mircosoft Office Suite, ImajeJ

Soft Skills

- Quick Learner, Adaptive

HOBBIES

Badminton, Cooking, Running